FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*
   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on* this section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

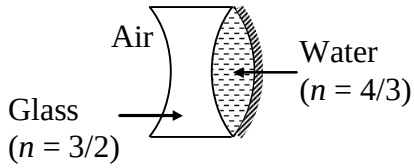
Physics

PART – I

SECTION – A
(One or More than one correct type)

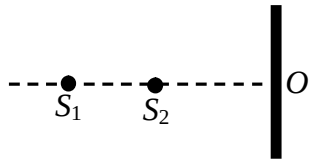
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. The radius of curvature of the left and right surface of the concave lens are 10 cm and 15 cm respectively. The radius of curvature of the mirror is 15 cm



(A) equivalent focal length of the combination is –18 cm
(B) equivalent focal length of the combination is +36 cm
(C) the system behaves like a concave mirror
(D) the system behaves like a convex mirror

2. Two mohochromatic and coherent point sources of light of wavelength λ are placed on the dotted line in front of an finite screen. The source emit waves in phase with each other. The distance between S_1 and S_2 is d while their distance from the screen is much larger. Then

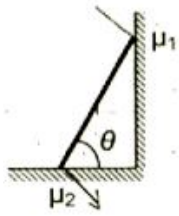


(A) If $d = 7\lambda/2$, O will be a minima
(B) If $d = 4.3\ \lambda$, there will be a total of 8 minima on screen
(C) If $d = 7\ \lambda$, O will be a maxima
(D) If $d = \lambda$, there will be only one maxima on the screen

3. A wave disturbance in a medium is described by
$y(x, t) = 0.02 \cos (50\ \pi t + \pi/2) \cos (10\ \pi x)$,
where 'x' and 'y' are in metre and 't' in seconds.
(A) A node occurs at x = 0.15 m
(B) An antinode occurs at x = 0.3 m
(C) The speed of the component wave is 5.0 m/s
(D) The wavelength is 0.2 m.

Space for Rough work

4. In figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium making an angle θ with the horizontal floor. The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then



(A) $\mu_1 = 0 \;\; \mu_2 \neq 0$ and $N_2 \tan\theta = \dfrac{mg}{2}$

(B) $\mu_1 \neq 0 \;\; \mu_2 = 0$ and $N_1 \tan\theta = \dfrac{mg}{2}$

(C) $\mu_1 \neq 0 \;\; \mu_2 \neq 0$ and $N_2 = \dfrac{mg}{1+\mu_1\mu_2}$

(D) $\mu_1 = 0 \;\; \mu_2 \neq 0$ and $N_1 \tan\theta = \dfrac{mg}{2}$

5. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.

(A) It the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.

(B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\dfrac{30}{\sqrt{2}}$ V.

(C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.

(D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

6. An ammeter has a resistance of 50 Ω and a full scale deflection current of 50 μA. It can be used as a voltmeter or as a higher range ammeter provided that a resistance is added to it. Choose the correct range and resistance combination from the following

(A) 20 V range with approximately 400 kΩ resistance in series

(B) 100 V range with approximately $2 \times 10^6 \Omega$ resistance in series

(C) 1 mA range with 50 Ω resistance in parallel

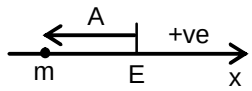
(D) 0.1 mA range with 50 Ω resistance in parallel

Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8



A particle of mass m is constrained to move on x-axis. A fore F acts on the particle. F always points toward the position labeled E. For example, when the particle is to the left of E, F points to the right. The magnitude of F is a constant F except at point E where it is zero

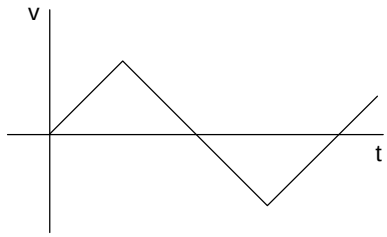
The system is horizontal. F is the net force acing on the particle. The particle is displaced a distance A towards left from the equilibrium position E and released from rest at t = 0.

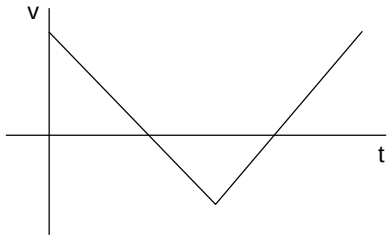
7. What is the period of the motion?

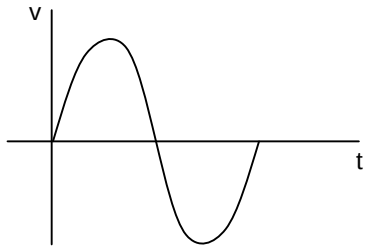
(A) $4\left(\sqrt{\frac{2Am}{F}}\right)$ (B) $2\left(\sqrt{\frac{2Am}{F}}\right)$

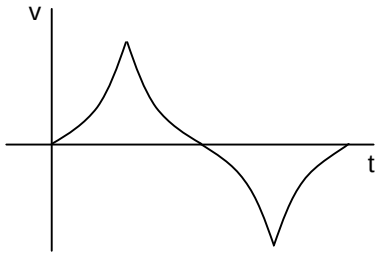
(C) $\left(\sqrt{\frac{2Am}{F}}\right)$ (D) None of these

8. Velocity–time graph of the particle is

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Paragraph for Question Nos. 9 and 10

The main scale of a vernier callipers reads in millimetre and its vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws of the instrument touch each other the seventh division of the vernier scale coincide with a scale division and the zero of the vernier lies to the right of the zero of main scale. Furthermore, when a cylinder is tightly placed along its length between the two jaws, the zero of the vernier scale lies slightly to the left of 3.2 cm; and the fourth vernier division coincides with a scale division. Calculate the measured length of the cylinder.

9. The zero error in the apparatus is
(A) –0.07 cm (B) 0.07 cm
(C) 0.03 cm (D) –0.03 cm

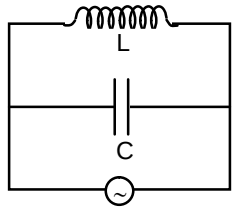
10. The measured value of the length of the cylinder is
(A) 3.14 cm (B) 3.24 cm
(C) 3.07 cm (D) 3.17 cm

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A non-viscous liquid of constant density 10^3 kg/m^3 flows in streamline motion along a vertical tube PQ of variable cross-section. Height of P and Q are 2m and 2.5m respectively. Area of tube at Q = 3 time area of tube at P. Find the work done per unit volume (in J/m^3) by pressure as liquid flows from P to Q. Speed of liquid at P is 3 m/s ($g = 10$ m/s^2).

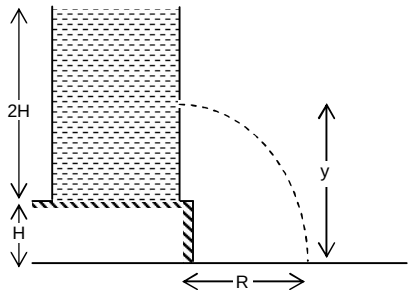
12. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. Find the current (in A) drawn from the generator



Generator

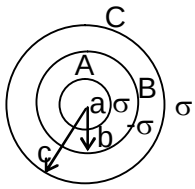
Space for Rough work

13. A tank is filled upto a height 2H with a liquid and is placed on a platform of height H from the ground. The distance y (in m) from the ground where a small hole is made in the tank, to get the maximum horizontal range R is (take H = 1.5 m)



14. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec. (Take 4kQ = 2.50 S.I. unit and R = 1m)

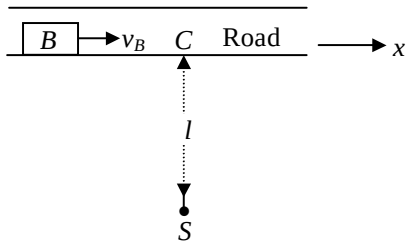
15. The adjacent figure shows charged spherical shells A, B and C having charge densities σ,-σ, σ and radii a, b, c respectively. If $V_A=V_c$ then c is equal to (in m) (assuming a = 0.10 m, b = 0.20 m)



16. A string of length 1 m and mass 10 gm is tightly clamped at its ends. The tension in the string is 4 N. Identical wave pulses are produced at one end at equal intervals of time t. What is the minimum value of t (in sec.) which allows a constructive interference between successive pulses?

17. A cubical body floats in a mercury bath with half of its volume submerged. What fraction of the body will be inside mercury if a layer of water is poured on the mercury covers the body completely? (Sp. gr. of mercury = 13.6)

18. A bus B is moving with a velocity v_B in the positive x-direction along a road as shown in the figure. A shooter 'S ' is at a distance l from the road. He has a detector which can detect signals only of frequency 1500Hz. The bus blows horn of frequency 1000 Hz. When the detector detects a signal the shooter immediately shoots towards the road along SC and the bullet hits the bus. Find the velocity of the bullet if velocity of sound in air is v = 340 m/s and $\frac{v_B}{v}=\frac{2}{3\sqrt{3}}$.



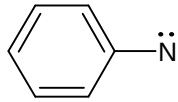
Space for Rough work

Chemistry

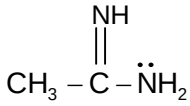
PART – II

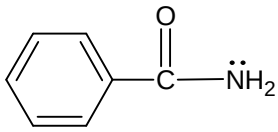
SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following compounds are more basic than  ?

(A) $CH_3 - CH_2 - \ddot{N}H_2$

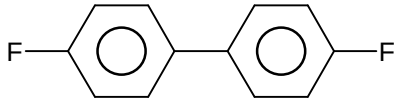
(B) 

(C) 

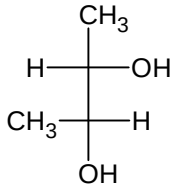
(D) $CH_2 = CH - \ddot{N}H_2$

20. Which of the following compounds would be optically inactive?

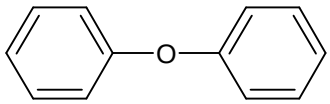
(A) $H_2C = C = CH_2$

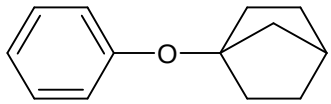
(B) 

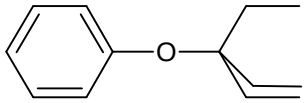
(C) $H_3C - CH = C = CH - CH_3$

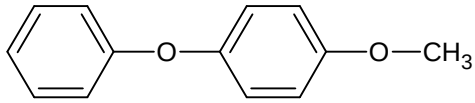
(D) 

21. Which of the following ethers will get hydrolysed by HI?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

22. Ionisation energy order is CORRECT for:
(A) $Sc^{3+} > Sc^{2+} > Sc^{+}$ (B) $Sc^{3+} > Ti^{4+} > V^{5+}$
(C) Sc > Y > La (D) Sc > Ca > K

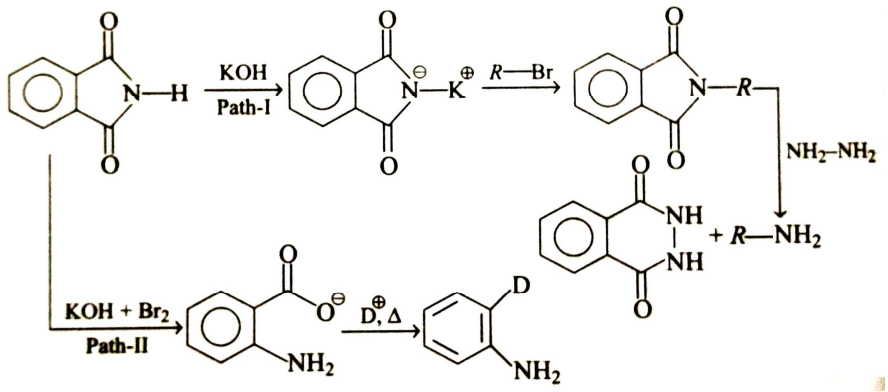
23. In which case EAN value is satisfied?
(A) $\left[Fe\left(\eta^5 C_5H_5\right)_2\right]$ (B) $\left[Ni\left(DMG\right)_2\right]$
(C) $\left[Mn_2\left(CO\right)_{10}\right]$ (D) $\left[Mn\left(CO\right)_5\right]^{1-}$

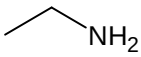
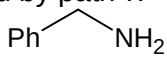
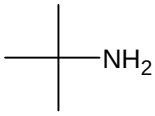
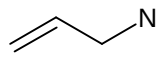
24. NaH produces H_2 gas when:
(A) it reacts with water (B) it reacts with heavy water
(C) it is electrolysed in fused state (D) it is heated for decomposition

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

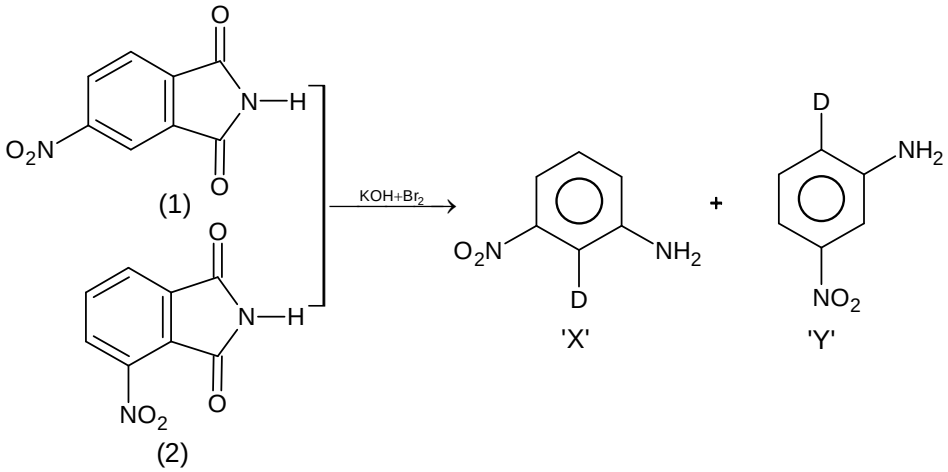
Paragraph for Question Nos. 25 and 26



25. Which of the following amines cannot be prepared by path-I?
(A)  (B) 
(C)  (D) 

Space for Rough work

26. Consider path II, choose the major product for 1 and 2.



(A) 1 – Y, 2 – X (B) 1 – X, 2 – Y
(C) 1 – X, 2 – X (D) 1 – Y, 2 – Y

Paragraph for Question Nos. 27 and 28

Bond length: Internuclear distance between two adjacent atoms in a species is known as bond length. Bond length depends on:
(i) Size of the atom involved in the bond formation
(ii) Size of the orbitals involved in the bond formation
(iii) Lone pair-lone pair repulsion
(iv) Resonance (also backbonding)
(v) s-character of combining orbital
With the increasing size of the atoms and atomic orbitals bond length increase. Lone pair-lone pair repulsion increases bond length (if atoms are small sized) whereas resonance can increase some bond lengths and decrease some other bond length. With increasing s-character bond length decreases, and with increasing multiplicity of bonds, bond length decreases. However, in some cases, bond lengths are also affected by relative position of bonds (between two similar atoms). Usually but not always with increasing bond length, bond strength (and hence bond dissociation energy) decreases.

27. The correct order of B – F bond length follows the sequence:

(A) $BF_3 < BF_2OH < BF_2NH_2 < BF_4^-$ (B) $BF_2NH_2 < BF_2OH < BF_3 < BF_4^-$
(C) $BF_3 < BF_4^- < BF_2OH < BF_2NH_2$ (D) $BF_3 < BF_2NH_2 < BF_2OH < BF_4^-$

Space for Rough work

28. In which of the following all bonds are not equivalent?

(A) N_2O (B) CN_2^{2-}

(C) N_3^- (D) NO_2^-

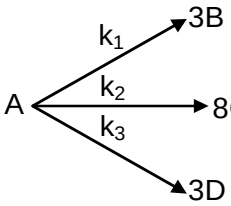
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. If edge fraction unoccupied in ideal FCC structure is x. Calculate the value of 10 x.

30. Calculate depression of freezing point for 0.66 molal aq. Solution of KCl.
(Given : $K_{f(H_2O)} = 1.86\ K\ kg\ mol^{-1}$)

31. If K_C for the reaction $Cu^{2+}(aq) + Sn^{2+}(aq) \longrightarrow Sn^{4+}(aq) + Cu(s)$ at 25°C is represented as $a \times 10^6$ then find the value of a. (Given : $E^o_{Cu^{2+}|Cu} = 0.34\ V$; $E^o_{Sn^{4+}|Sn^{2+}} = 0.15\ V$)

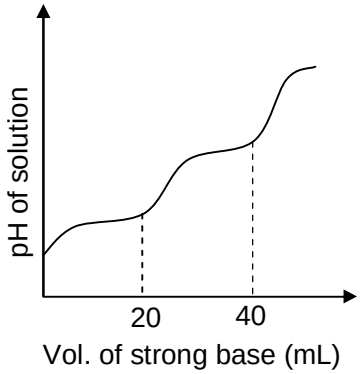
32. 

; At time t = 0, initial mole of A is 1.

Overall half life of the reaction is 15 days. Then calculate the number of mole of B + C + D after 45 days if the ratio of $k_1 : k_2 : k_3$ is 4 : 2 : 1.

Space for Rough work

33. 10 mL of H_2A (weak diprotic acid) solution is titrated against 0.1 M NaOH. pH of the solution is plotted against volume of strong base added and following observation is made.

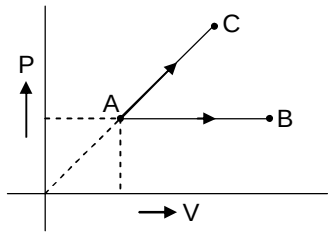


If pH of the solution at first equivalence point is pH_1 and at second equivalence point is pH_2.

Calculate the value of $\frac{pH_2}{pH_1}$ at 25°C

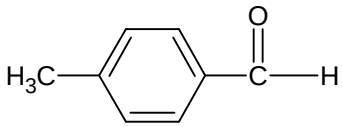
Given : For H_2A, $pK_{a_1} = 4.6$ and $pK_{a_2} = 8$, $\log 25 = 1.4$

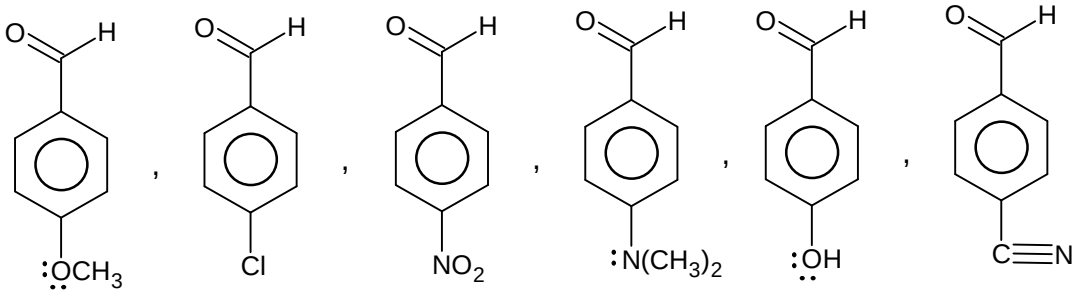
34. One mole ideal monoatomic gas is heated according to path AB and AC.
If temperature of state B and state C are equal.
Calculate $\frac{q_{AC}}{q_{AB}}$.



Space for Rough work

35. Examine the structural formulas of following compounds and find out number of compounds which show higher rate of nucleophilic addition than





36. Total number of substance which contain hexagonal planer rings in their structures (graphite, $(BN)_x$, $B_3N_3H_6$, C_6H_6, B_2H_6, $H_3P_3O_9$):

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $S_2 = 0$ be the mirror image of $S_1 : x^2 + y^2 - 4x - 6y + 12 = 0$ w.r.t. the line $L_1 : 10^4 x + \left(10^4 + 10\right)y + \left(10^4 + 20\right) = 0$. Let $L_2 : 2^{11}x + \left(2^{11} + 2^{12}\right)y + \left(2^{11} + 2^{13}\right) = 0$ be a line. The equation of the line passing through the point of intersection of the line $L_2 = 0$ with the radical axis of $S_1 = 0, S_2 = 0$ and making equal intercepts with the coordinate axes, is

(A) $x - y - 3 = 0$ (B) $x + y + 1 = 0$

(C) $2x - 2y + 1 = 0$ (D) $2x + 2y + 3 = 0$

38. A chord of negative slope from the point $P\left(\sqrt{264}, 0\right)$ is drawn to the ellipse $x^2 + 4y^2 = 16$. This chord intersects the ellipse at A and B (O is the origin).

(A) The maximum area of ΔAOB is 4

(B) The maximum area of ΔAOB is 8

(C) If the area of ΔAOB is maximum, then the slope of line AB is $\frac{-1}{2\sqrt{2}}$

(D) If the area of ΔAOB is maximum, then the slope of line AB is $\frac{-1}{8\sqrt{2}}$

39. Equation of plane passing through the line $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-2}{-2}$ and making an angle of 30° with the plane x + y + z = 5 is.

(A) $(x - y - 1) + \left(2 - \sqrt{2}\right)(2y + z - 6) = 0$

(B) $(x - y + 1) + \left(2 + \sqrt{2}\right)(2y + z - 6) = 0$

(C) $(x - y - 1) + \left(2 + \sqrt{2}\right)(2y + z - 6) = 0$

(D) $(x - y + 1) + \left(2 - \sqrt{2}\right)(2y + z - 6) = 0$

Space for Rough work

40. The volume of the parallelepiped whose coterminous edges are represented by the vectors $2\vec{b}\times\vec{c}$, $3\vec{c}\times\vec{a}$ and $4\vec{a}\times\vec{b}$, where $\vec{a}=(1+\sin\theta)\hat{i}+\cos\theta\hat{j}+\sin 2\theta\hat{k}$,
$\hat{b}=\sin\left(\theta+\frac{2\pi}{3}\right)\hat{i}+\cos\left(\theta+\frac{2\pi}{3}\right)\hat{j}+\sin\left(2\theta+\frac{4\pi}{3}\right)\hat{k}$,
$\vec{c}=\sin\left(\theta-\frac{2\pi}{3}\right)\hat{i}+\cos\left(\theta-\frac{2\pi}{3}\right)\hat{j}+\sin\left(2\theta-\frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then the value of θ, in the interval $\left(0,\frac{\pi}{2}\right)$, is are

(A) $\frac{\pi}{9}$ (B) $\frac{2\pi}{9}$

(C) $\frac{\pi}{3}$ (D) $\frac{4\pi}{3}$

41. A player tosses a coin. He sets one point for head and two points for tail. He plays till he gets sum of points equal to n. If p_n be the probability that his score becomes n, then

(A) $p_3=\frac{1}{2}$ (B) $p_n=\frac{1}{2}P_{n-1}+\frac{1}{4}P_{n-2}$

(C) $p_4=\frac{11}{16}$ (D) $p_n=\frac{1}{2}(p_{n-1}+p_{n-2})$

42. If the imaginary part of the complex number
$(z-1)(\cos\alpha-i\sin\alpha)+(z-1)^{-1}(\cos\alpha+i\sin\alpha)$ is zero, then

(A) $|z|=1$ (B) $|z-1|=1$

(C) $\arg(z)=\alpha$ (D) $\arg(z-1)=\alpha$

Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Consider the function $f(x) = b\log_e x - x$ on the interval $(0, \infty)$, where b is a positive real constant.

43. The set of all values of b for which the equation $b\log x - x = 0$ has exactly one real roots is equal to

(A) $\left(0, \frac{1}{e}\right)$ (B) (0, 1)

(C) $(-\infty, 0)$ (D) none of these

44. If the line $x - y = 0$ is tangent to $f(x) = b\log_e x - x,$ then b lies in the interval

(A) (1, 3) (B) (0, 1)

(C) (4, 6) (D) (5, 7)

Paragraph for Question Nos. 45 and 46

Given a ΔABC in which AD is perpendicular to base BC, Incircle with radii r_1 and r_2 are drawn in ΔABD and ΔACD respectively. They touch the altitude AD at point M and N respectively. Assume sides and angles have usual meaning.

45. If ΔABC is an isosceles triangle and the angle between the equal sides AB and AC is $\frac{2\pi}{3}$, then $r_1 : r_2$ is

(A) $\sqrt{3}+1 : \sqrt{3}-1$ (B) $3+\sqrt{3} : 2-\sqrt{3}$

(C) $3-\sqrt{3} : 3$ (D) none of these

Space for Rough work

46. If R be the circum radius of ΔABC then MN will be

(A) $\left|2R\sin\frac{B-C}{2}.\cos\frac{B}{2}\cos\frac{C}{2}\right|$ (B) $\left|4R\sin\left(\frac{C-A}{2}\right)\cos\frac{B}{2}\sin\frac{C}{2}\right|$

(C) $\left|4R\sin\left(\frac{B-C}{2}\right)\sin\frac{B}{2}\sin\frac{C}{2}\right|$ (D) None of these

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Let $f(x)=x^3+3x+2$ and $g(x)$ be its inverse. If the area bounded by $g(x)$, x – axis and the ordinates x = – 2 and x = 6 is $\frac{p}{q}$ (where p& q are coprime), find value of $p+q-1$

48. If the maximum value of $\left|\sqrt{x^4-7x^2-4x+20}-\sqrt{x^4+9x^2+16}\right|$ is P, find the value of $[P]$ (where [.] denotes GIF)

49. If $\int_0^{\pi} x\,f\left(\cos^2 x+\tan^4 x\right)dx=\pi k\int_0^{\pi/2} f\left(\cos^2 x+\tan^4 x\right)dx$, then the value of k is

50. If $a,b,c\in R^+$ and $a^2bc+4abc+4bc=81$. Find the value of $a+b+c$ such that $2a+b+c+4$ assumes its least value _____

51. Let $A=\begin{bmatrix}2 & 0\\ 0 & -3\end{bmatrix}$, $B=\begin{bmatrix}0 & 1\\ 3 & 0\end{bmatrix}$ and $x=\begin{bmatrix}\cos x & -\sin x\\ \sin x & \cos x\end{bmatrix}$. If $P=AXB$, $Q=BX^TA$ and $Tr.\left((PQ)^{10}\right)=a^{10}+b^{10}$, where $a<b$, then find the value of $b-2a$ _____

(Tr (M) denotes trace of Matrix M and M^T denotes transpose of matrix M)

Space for Rough work

52. Let $1+\sum_{r=1}^{10}\left(3^r.{}^{10}C_r + r.{}^{10}C_r\right) = 2^{10}\left(\alpha.4^5+\beta\right)$ where $\alpha,\beta \in N$ and $f(x) = x^2 - 2x - k^2 + 1$.

If α,β lies between the roots of $f(x) = 0$, then find the smallest positive integral value of k.

53. If the straight line $x = 1+s,\ y = 3-\lambda s,\ z = 1+\lambda s$ and $x = \frac{t}{2},\ y = 1+t,\ z = 2-t$ with parameters s and t respectively, are coplanar, then the value of $|\lambda|$ is _____

54. If the sum of the first 3n terms is equal to the next n terms of an A.P. whose common difference is non – zero, then the reciprocal of the ratio of the sum of the first 2n terms to the next 2n terms is _____

Space for Rough work